

*For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only OR
for Specialist Use only*

Gefiticip 250

Gefitinib Tablets 250 mg

COMPOSITION

Each film coated tablet contains:

Gefitinib BP/Ph. Eur ...250 mg

Colour – Titanium dioxide, Iron Oxide Red and Iron oxide Yellow

Excipient with known effect: Lactose

DOSAGE FORM

Tablet

PRODUCT DESCRIPTION

Round, biconvex, brown, film coated tablets debossed with “C” on one side and plain on the other side.

PHARMACOLOGY

Pharmacodynamics

ATC Classification

L01XE02 - gefitinib; Belongs to the class of protein kinase inhibitors, other antineoplastic agents. Used in the treatment of cancer.

Gefitinib is a selective inhibitor of the epidermal growth factor receptor (EGFR) tyrosine kinase, commonly expressed in solid human tumours of epithelial origin. Inhibition of EGFR tyrosine kinase activity inhibits tumour growth, metastasis and angiogenesis and increases tumour cell apoptosis.

Patients that have never smoked, have adenocarcinoma histology, are female gender or are of Asian ethnicity, are more like to benefit from treatment with gefitinib. These clinical characteristics are also associated with a higher rate of EGFR mutation positive tumours.

Pharmacokinetics

Following intravenous administration, gefitinib is rapidly cleared, extensively distributed and has a mean elimination half-life of 48 hours. Following oral dosing in cancer patients, absorption is moderately slow and the mean terminal half-life is 41 hours. Administration of gefitinib once daily results in 2 to 8-fold accumulation with steady state exposures achieved after 7 to 10 doses. At steady state, circulating plasma concentrations are typically maintained within a 2 to 3-fold range over the 24-hour dosing interval.

Absorption

Following oral administration of gefitinib, peak plasma concentrations of gefitinib typically occur at 3 to 7 hours after dosing. Mean absolute bioavailability is 59% in cancer patients. Exposure to gefitinib is not significantly altered by food. Where gastric pH was maintained above pH 5, gefitinib exposure was reduced by 47%.

Distribution

Mean volume of distribution at steady state of gefitinib is 1400 L indicating extensive distribution into tissue. Plasma protein binding is approximately 90%. Gefitinib binds to serum albumin and α 1-acid glycoprotein.

Metabolism

CYP3A4 is the major P450 isozyme involved in the oxidative metabolism of gefitinib.

Gefitinib has limited potential to inhibit CYP2D6. Gefitinib was co-administered with metoprolol (a CYP2D6 substrate). This resulted in a small increase in exposure to metoprolol, which is not considered to be clinically relevant.

Gefitinib shows no enzyme induction effects in animal studies and no significant inhibition of any other cytochrome P450 enzyme.

Three sites of biotransformation have been identified in the metabolism of gefitinib: Metabolism of the N-propylmorpholino-group, demethylation of the methoxy- substituent on the quinazoline and oxidative defluorination of the halogenated phenyl group. Five metabolites have been fully identified in faecal extracts and the major component was O-desmethyl gefitinib, although this only accounted for 14% of the dose.

In human plasma 8 metabolites were fully identified. The major metabolite identified was O-desmethyl gefitinib, which was 14-fold less potent than gefitinib at inhibiting EGFR stimulated cell growth and had no inhibitory effect on tumour cell growth in mice. It is therefore considered unlikely that it contributes to the clinical activity of gefitinib.

The production of O-desmethyl gefitinib has been shown to be via CYP2D6.

In poor metabolisers, no measurable levels of O-desmethyl gefitinib were produced. The range of gefitinib exposures achieved in both the extensive and the poor metaboliser groups were wide and overlapping but the mean exposure to gefitinib was 2-fold higher in the poor metaboliser group. The higher average exposures that could be achieved by individuals with no active CYP2D6 may be clinically relevant since adverse experiences are related to dose and exposure.

Elimination

Gefitinib total plasma clearance is approximately 500 mL/min. Excretion is predominantly via the faeces with renal elimination of drug and metabolites accounting for less than 4% of the administered dose.

Special Populations

Cancer patients, no relationships were identified between predicted steady state trough concentration and patient age, body weight, gender, ethnicity or creatinine clearance.

Single dose gefitinib 250 mg in patients with mild, moderate or severe hepatic impairment due to cirrhosis (according to Child-Pugh classification), there was an increase in exposure. An average 3.1-fold increase in exposure to gefitinib in patients with moderate and severe hepatic impairment was observed. None of the patients had cancer, all had cirrhosis and some had hepatitis. This increase in exposure may be of clinical relevance since adverse experiences are related to dose.

INDICATIONS

Gefticip 250 is indicated for the treatment of adult patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with activating mutations of EGFR-TK.

DOSAGE AND METHOD OF ADMINISTRATION

The recommended dose of gefitinib is one 250 mg tablet once a day, taken with or without food. If a dose of gefitinib is missed, it should be taken as soon as the patient remembers. If it is less than 12 hours to the next dose, the patient should not take the missed dose. Patients should not take a double dose (two doses at the same time) to make up for a forgotten dose.

Gefitinib is not recommended for use in children or adolescents as safety and effectiveness in these patient populations has not been studied.

No dosage adjustment is required on the basis of patient age, body weight, gender, ethnicity or renal function or in patients with moderate to severe hepatic impairment due to liver metastases.

Dosage Adjustment: Patients with poorly tolerated diarrhoea or skin adverse drug reactions may be successfully managed by providing a brief (up to 14 days) therapy interruption followed by reinstatement of the 250 mg dose (see *Undesirable effects*).

Administration

Where dosing of whole tablets is not possible, such as patients who are only able to swallow liquids, tablets may be administered as a dispersion in water. The tablet should be dropped into half a glass of drinking water (non-carbonated), without crushing, and the glass is stirred until the tablet has dispersed (approximately 15 minutes) and the contents subsequently drunk immediately. The glass should be rinsed with a further half glass of water and the contents drunk.

The liquid can also be administered via a nasogastric tube.

CONTRAINDICATIONS

Known severe hypersensitivity to the active substance or to any of the excipients of this product.

WARNINGS AND PRECAUTIONS

When considering the use of gefitinib as first-line treatment for locally advanced or metastatic NSCLC, it is recommended that EGFR mutation assessment of the tumour tissue is attempted for all patients. When assessing the mutation status of a patient it is important that a well-validated and robust methodology is chosen to minimise the possibility of false negative or false positive determinations. Tumour samples which are used for the diagnosis of advanced NSCLC are the preferred sample type for EGFR mutation testing. A tumour sample should be collected and tested where possible. If a tumour sample is not available or evaluable, then circulating tumour DNA (ctDNA) obtained from a blood (plasma) sample may be used. Only robust, reliable, sensitive test(s) with demonstrated utility on ctDNA should be used for the determination of EGFR mutation status of ctDNA. EGFR mutations identified in ctDNA are highly predictive of EGFR mutation positive tumours). However it is not always possible to detect EGFR mutations using this sample type (0.2% false positives, 34.3% false negatives).

In the first line setting, gefitinib should not be used in preference to doublet chemotherapy in mutation-negative patients.

Interstitial Lung Disease (ILD), which may be acute in onset, has been observed in patients receiving gefitinib, and some cases have been fatal (see *Undesirable effects*). If patients present with worsening of respiratory symptoms such as dyspnoea, cough and fever, gefitinib should be interrupted and prompt investigation initiated. If ILD is confirmed, gefitinib should be discontinued and the patient treated appropriately.

In patients with NSCLC who were receiving gefitinib or chemotherapy, the following risk factors for developing ILD (irrespective of whether the patient received gefitinib or chemotherapy) were identified:

Smoking, poor performance status (PS ≥ 2), CT scan evidence of reduced normal lung ($\leq 50\%$), recent diagnosis of NSCLC (< 6 months), pre-existing ILD, increasing age (≥ 55 years old) and concurrent cardiac disease. Risk of mortality among patients who developed ILD on both treatments was higher in patients with the following risk factors: Smoking, CT scan evidence of reduced normal lung ($\leq 50\%$), pre-existing ILD, increasing age (≥ 65 years old), and extensive areas adherent to pleura ($\geq 50\%$).

Liver function test abnormalities (including increase in alanine aminotransferase, aspartate aminotransferase, bilirubin) have been observed (see *Undesirable effects*), uncommonly presenting as hepatitis. There have been isolated reports of hepatic failure which in some cases led to fatal outcomes. Therefore, periodic liver function testing is recommended. gefitinib should

be used cautiously in the presence of mild to moderate change in liver function. Discontinuation should be considered if changes are severe.

Cerebrovascular events have been reported. A relationship with gefitinib has not been established.

Substances that are inducers of CYP3A4 activity may increase metabolism and decrease gefitinib plasma concentrations. Therefore, co-medication with CYP3A4 inducers (e.g. phenytoin, carbamazepine, rifampicin, barbiturates or St John's Wort) may reduce efficacy (see *Drug Interactions*).

International Normalised Ratio (INR) elevations and/or bleeding events have been reported in some patients taking warfarin (see *Drug Interactions*). Patients taking warfarin should be monitored regularly for changes in Prothrombin Time (PT) or INR.

Drugs that cause significant sustained elevation in gastric pH may reduce plasma concentrations of gefitinib and therefore may reduce efficacy (see *Pharmacokinetics and Drug Interactions*).

Patients should be advised to seek medical advice promptly in the event of developing: Severe or persistent diarrhoea, nausea, vomiting or anorexia.

These symptoms should be managed as clinically indicated (see *Undesirable effects*). Patients presenting with signs and symptoms suggestive of keratitis such as acute or worsening: Eye inflammation, lacrimation, light sensitivity, blurred vision, eye pain and/or red eye should be referred promptly to an ophthalmology specialist.

If a diagnosis of ulcerative keratitis is confirmed, treatment with gefitinib should be interrupted, and if symptoms do not resolve, or recur on reintroduction of gefitinib, permanent discontinuation should be considered.

An increased risk of cerebral haemorrhage in adult patients with NSCLC receiving gefitinib has not been established.

Gefitinib and vinorelbine have been used concomitantly, indicate that gefitinib may exacerbate the neutropenic effect of vinorelbine.

Gastrointestinal perforation has been reported in patients taking gefitinib. In most cases this is associated with other known risk factors, including increasing age, concomitant medications such as steroids or NSAIDs, underlying history of GI ulceration, smoking or bowel metastases at sites of perforation.

Lactose intolerance

This medicine contains lactose. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

DRUG INTERACTIONS

Metabolism of gefitinib is predominantly via CYP3A4.

Co-administration with rifampicin (a known potent CYP3A4 inducer) reduced mean gefitinib AUC by 83% of that without rifampicin.

Co-administration with itraconazole (a CYP3A4 inhibitor) resulted in an 80% increase in the mean AUC of gefitinib. This increase may be clinically relevant since adverse experiences are related to dose and exposure.

Co-administration of ranitidine at a dose that caused sustained elevations in gastric pH (≥ 5), resulted in a reduced mean gefitinib AUC by 47%.

INR elevations and/or bleeding events have been reported in some patients taking warfarin.

FERTILITY, PREGNANCY AND LACTATION

There are no data from the use of gefitinib in pregnant or breast-feeding women.

Studies in animals have shown reproductive toxicity. Animal studies also indicate that gefitinib and certain metabolites pass into rats breast-milk

Women of childbearing potential must be advised to avoid becoming pregnant, and breast-feeding mothers must be recommended to discontinue nursing while receiving gefitinib therapy.

UNDESIRABLE EFFECTS

The most commonly reported adverse drug reactions (ADRs) are diarrhea, and skin reactions (including rash, acne, dry skin and pruritus). ADRs usually occur within the first month of therapy and are generally reversible.

Frequency of ADRs relating to abnormal laboratory values is based on patients with a 2 or more CTC grade change from baseline in the relevant laboratory parameters. (See Table 1).

Table 1: Adverse Drug Reactions by Frequency and System / Organ		
Very common	Gastrointestinal disorders	Diarrhoea, mainly mild or moderate (CTC grade 1 or 2) and, less commonly, severe (CTC grade 3 or 4). Nausea, mainly mild (CTC grade 1). Vomiting, mainly mild or moderate (CTC grade 1 or 2). Stomatitis, predominantly mild (CTC grade 1).
	Hepatobiliary disorders	Elevations in alanine aminotransferase, mainly mild to moderate
	Metabolism and nutrition disorders	Anorexia, mild or moderate (CTC grade 1 or 2)
	Skin and subcutaneous tissue disorders	Skin reactions, mainly a mild or moderate (CTC grade 1 or 2) pustular rash, sometimes itchy with dry skin, including skin fissures on an erythematous base.
	General Disorders and administration site conditions	Asthenia, predominantly mild (CTC grade 1)
Common	Gastrointestinal disorders	Dehydration, secondary to diarrhea, nausea, vomiting or anorexia. Dry mouth, predominantly mild (CTC grade 1)
	Vascular disorders	Haemorrhage, such as epistaxis and haematuria
	Hepatobiliary disorders	Elevations in Aspartate aminotransferase, mainly mild to moderate. Elevations in total bilirubin, mainly mild to moderate
	Renal and urinary disorders	Asymptomatic laboratory elevations in blood creatinine Proteinuria Cystitis
	Skin and subcutaneous tissue disorders	Nail disorder Alopecia
	General disorders and administration site conditions	Pyrexia
	Eye disorders	Conjunctivitis, blepharitis, and dry eye, mainly mild (CTC grade 1)
	Respiratory, thoracic and mediastinal disorders	Interstitial lung disease, often severe (CTC grade 3-4). Fatal outcomes have been reported.
	Skin and subcutaneous tissue disorders	Allergic reactions, including angioderma and urticaria

Uncommon	Gastrointestinal disorders	Pancreatitis Gastrointestinal perforation
	Hepatobiliary disorders	Hepatitis
	Eye disorders	Keratitis Corneal erosion, reversible and sometimes in association with aberrant eyelash growth
Rare	Skin and subcutaneous tissue disorders	Bullous conditions including Toxic epidermal necrolysis, Stevens-Johnson syndrome and erythema multiforme Cutaneous vasculitis
	Renal and urinary disorders	Haemorrhagic cystitis

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the local reporting system.

OVERDOSE

There is no specific treatment in the event of overdose of gefitinib. Adverse reactions associated with overdose should be treated symptomatically; in particular severe diarrhoea should be managed as clinically indicated.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

During treatment with gefitinib, asthenia has been reported and those patients who experience this symptom should observe caution when driving or using machines.

STORAGE CONDITION

Store below 30°C

SHELF LIFE

24 months

PACKAGING INFORMATION

Three blister packs (Clear PVC / Plain ALU Blister) containing 10 tablets each wrapped in aluminum foil pouch.

MANUFACTURED BY

Cipla Ltd

S-103 to S-105, S-107 to S-112,

L-138, L-147, L-147/1 to L-147/3 & L-147/A

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PRODUCT REGISTRATION HOLDER

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DATE OF REVISION

April 2021